POWER BI PROJECT REPORT

August – January 2026

Global Recession and Layoff Tracker

Submitted by

Diwakar Singh

Registration No - 12307725

Programme -P132, Section – K23MS

Course Code - INT374

Under the Guidance of

Savleen Kaur (18306)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Diwakar Singh**, bearing Registration No. **12307725**, has successfully completed the INT374 project titled **"India City GDP Analysis using Power BI"** under my guidance and supervision. To the best of my knowledge, this project is the result of his original work, effort, and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University Phagwara, Punjab.

Date: 22/12/2025

DECLARATION

I, **Diwakar Singh**, student of B.Tech Computer Science and Engineering under the CSE/IT discipline at Lovely Professional University, Punjab, hereby declare that the work presented in this project report is my original work and has not been submitted elsewhere for any academic purpose.

Date: 22/12/2025 Diwakar

Registration No. 12307725 Diwakar Singh

ACKNOWLEDGEMENT

The successful completion of this project, " **India City GDP Analysis using Power BI**", would not have been possible without the guidance and support of several individuals and resources.

First and foremost, I would like to express my deepest gratitude to my project mentor and professor, [Savleen Kaur mam], for her invaluable guidance, constant encouragement, and constructive feedback throughout the duration of this project. Her insights into data analytics and business intelligence helped shape the direction of this analysis and overcome various technical challenges.

I am also grateful to the School of Computer Science and Engineering at Lovely Professional University for providing the necessary infrastructure and academic environment to carry out this work. I would like to extend my thanks to the Head of the Department and the faculty members for their support.

I must also acknowledge the open-data community and the providers of the datasets used in this analysis. Access to this high-quality, real-world data was essential for the creation of the Power BI dashboards presented in this report.

Diwakar Singh

Registration No. – 12307725

Date – 22/12/2025

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1. Introduction

1.1 Background: The Volatile Economic Landscape (2020–2024)

India's economy functions as a large, diverse, and dynamic system where macroeconomic indicators such as Gross Domestic Product (GDP), sectoral output, employment, and urban economic contribution act as key measures of national economic health. Over the past decade, India has experienced significant structural and cyclical changes, positioning itself as one of the fastest-growing major economies in the world. The period from 2015 to 2024, in particular, represents a crucial phase marked by rapid urbanization, digital transformation, policy reforms, and external economic shocks.

A defining event during this period was the COVID-19 pandemic, which caused a sharp contraction in economic activity during 2020. Nationwide lockdowns disrupted supply chains, reduced industrial output, and severely impacted service-driven urban economies. However, the recovery phase that followed was uneven across regions and sectors. While certain metropolitan cities and digital-driven industries such as Information Technology, financial services, and telecommunications rebounded strongly, other sectors such as tourism, manufacturing, and small-scale enterprises experienced prolonged stress.

Post-2021, India's economic recovery was influenced by rising global inflation, energy price volatility, and tightening monetary conditions. Despite these challenges, urban centers continued to play a dominant role in driving GDP growth, highlighting the growing importance of city-level economic analysis. This divergence between cities, sectors, and employment patterns underscores the need for granular, data-driven evaluation beyond national GDP aggregates.

1.2 The Importance of City-Level and Sectoral Economic Analysis

Traditional economic analysis in India has largely focused on national or state-level indicators, which often fail to capture disparities across cities and industries. Major metropolitan regions such as Mumbai, Delhi, Bengaluru, Chennai, and Hyderabad contribute a disproportionate share to India's GDP due to concentrated industrial activity, skilled labor availability, and infrastructure development. At the same time, sectoral performance varies significantly across regions, influencing employment generation and income distribution.

Understanding the relationship between GDP growth, sectoral contribution, and employment trends is essential for effective policymaking and strategic planning. For instance, growth in the IT and services sector has led to increased urban employment, whereas stagnation in manufacturing or MSMEs directly affects job creation. Analyzing these patterns helps stakeholders identify high-growth regions, underperforming sectors, and emerging economic hubs.

However, such analysis becomes challenging when data is scattered across multiple government sources and presented in static formats. Without interactive tools, identifying correlations—such as how sectoral growth impacts employment or how city-level GDP influences national performance—remains complex and time-consuming.

1.3 Problem Statement

Despite the availability of extensive economic data from government and authorized sources, there remains a lack of accessible and integrated analytical tools that enable a comprehensive analysis of India's GDP performance at national, city, and sectoral levels. Conventional reporting methods often fail to answer critical economic questions such as:

- How do different Indian cities contribute to national GDP, and how does this contribution vary over time?

- What is the relationship between sectoral growth (IT, manufacturing, services, tourism, MSMEs) and overall economic performance?
- How does GDP growth influence employment and unemployment trends across major urban centers?

Most existing analyses rely on static reports, spreadsheets, or isolated charts, which limit the ability to explore economic patterns dynamically. For example, GDP figures may be presented in one document, while employment statistics are shown in another, requiring manual interpretation and increasing the likelihood of analytical bias or oversight.

Additionally, traditional tools do not allow users to interactively compare cities, sectors, and time periods, making it difficult to identify high-growth regions, underperforming sectors, or emerging economic hubs. As a result, valuable insights into India's urban economic structure remain underutilized.

Therefore, there is a pressing need for an interactive Business Intelligence (BI) solution that consolidates GDP, sectoral contribution, and employment data into a unified analytical framework. Such a system can provide a holistic, data-driven view of India's economic landscape and support informed decision-making for policymakers, researchers, and analysts.

1.4 Objectives of the Study

This project, **"**India GDP & City-Level Economic Analysis Dashboard**,"** aims to address existing gaps in traditional economic reporting by developing a comprehensive analytical framework using Microsoft Power BI. The primary objective of this study is to transform raw, fragmented economic datasets into an interactive and visually intuitive analytical platform that provides meaningful insights into India's economic performance.

Specifically, this report aims to:

1. **Consolidate Data Sources:**
   Design an efficient **Extract, Transform, Load (ETL)** process to integrate multiple authorized economic datasets, including GDP, sectoral contribution, employment, and unemployment data, into a unified and well-structured data model.
2. **Analyze GDP Trends Over Time:**
   Develop interactive visualizations to examine India's GDP growth patterns across

different years, enabling users to identify economic fluctuations and long-term growth trends.

3. **Evaluate City-Level Economic Contribution:**
   Perform a detailed analysis of city-wise GDP distribution to understand the economic dominance of major metropolitan regions and identify emerging urban growth centers.

4. **Assess Sectoral Performance:**
   Analyze the contribution of key economic sectors such as Information Technology, Manufacturing, Services, Tourism, and MSMEs to overall GDP, highlighting sector-wise strengths and vulnerabilities.

5. **Study Employment and Unemployment Patterns:**
   Explore the relationship between economic growth and employment indicators, providing insights into how sectoral and regional GDP performance influences job creation and unemployment levels.

6. **Enable Data-Driven Decision Making:**
   Deliver an interactive Business Intelligence dashboard that allows policymakers, researchers, and analysts to dynamically explore data, compare regions and sectors, and derive actionable economic insights.

2. Scope of the Analysis

The scope of this project is defined by the economic data domains**,** geographical coverage**,** and temporal range incorporated into the final analytical dashboard. The primary objective of this analysis is to examine India's economic performance by studying GDP trends, city-level contributions, sectoral distribution, and employment indicators using an interactive Business Intelligence approach.

Functional Scope

The analysis is organized into multiple thematic areas, corresponding to the interconnected Power BI dashboard pages developed in this project:

1. **National GDP Overview:**
   A comprehensive analysis of India's overall GDP trends across different years, focusing on growth patterns, economic fluctuations, and long-term performance.
2. **City-Level Economic Contribution:**
   An in-depth examination of city-wise GDP contribution, highlighting the economic dominance of major metropolitan regions and identifying emerging urban growth centers within India.
3. **Sectoral Contribution Analysis:**
   A detailed evaluation of key economic sectors such as Information Technology, Manufacturing, Services, Tourism, and MSMEs, analyzing their contribution to GDP at both national and city levels.
4. **Employment and Unemployment Analysis:**
   An assessment of employment and unemployment trends in relation to GDP growth, providing insights into how economic expansion or slowdown impacts job creation across regions and sectors.
5. **Comparative Growth Analysis:**
   A comparative study of cities and sectors over time to identify high-growth, stable, and underperforming segments of the Indian economy.

Data Scope

- **Geographical Coverage:**
  The analysis is focused on **India**, with special emphasis on major urban centers and economically significant cities.

- **Temporal Coverage:**
  The dataset spans multiple years, enabling trend analysis and year-over-year comparison of GDP, sectoral output, and employment indicators.
- **Key Metrics:**
  GDP, GDP Growth Rate, Sectoral Contribution, Employment Rate, Unemployment Rate, and City-wise Economic Share.

Target Audience

The insights derived from this analysis are intended for:

- Students and academic researchers
- Policy analysts and planners
- Business intelligence professionals
- Decision-makers seeking data-driven economic insights

3. Existing System

Currently, analyzing India's economic performance—particularly at the city and sectoral levels—is largely a manual and fragmented process for students, researchers, and analysts. Although economic data is available from multiple government and authorized sources, it is not presented in a consolidated or interactive manner.

The existing system generally involves the following steps:

1. **Disparate Data Collection:**
   Analysts must manually access multiple government portals and statistical reports to collect data related to GDP, sectoral output, employment, and unemployment. These datasets are often distributed across different websites and published in varying formats.
2. **Independent Economic Analysis:**
   City-wise GDP figures, sectoral statistics, and employment indicators are typically

published separately, requiring analysts to study each dataset in isolation without a unified analytical framework.

3. **Manual Data Processing using Spreadsheets:**
   The collected datasets are commonly downloaded in CSV or Excel formats and processed using spreadsheet tools such as Microsoft Excel. Functions like VLOOKUP, filtering, and manual aggregation are used to align GDP, sectoral, and employment data, making the process time-consuming and error-prone.

4. **Static and Non-Interactive Reporting:**
   The final analysis is usually presented through static tables or charts embedded in reports or presentations. These outputs lack interactivity, making it difficult to dynamically compare cities, sectors, or time periods or to perform drill-down analysis.

i. Drawbacks or limitations of existing system

The traditional approach to analyzing India's economic performance suffers from several significant limitations, which restrict effective interpretation and decision-making:

- **Lack of Scalability and Automation:**
  The manual process of collecting, cleaning, and merging economic datasets from multiple government sources is time-consuming and prone to human error. As new GDP, employment, and sectoral data is periodically released, spreadsheet-based analyses quickly become outdated.

- **Limited Multi-Dimensional Analysis:**
  Spreadsheet tools make it difficult to simultaneously analyze multiple economic dimensions—such as GDP growth, sectoral contribution, and employment trends—across cities and time periods, limiting deeper economic insight.

- **Difficulty in Identifying Relationships and Trends:**
  Visualizing correlations between economic indicators (for example, GDP growth versus employment levels across cities) is challenging using static charts, making it harder to identify patterns, comparisons, or regional disparities.

- **Absence of Interactive Exploration:**
  Static reports do not allow users to dynamically filter or drill down into data. For instance, if a user observes higher GDP growth in a particular year, they cannot

instantly analyze which cities or sectors contributed most to that growth without generating additional reports.

- **Data Silos:**
  Economic indicators such as GDP, sectoral output, and employment statistics are often analyzed in isolation, preventing a holistic understanding of India's economic structure.

This Power BI project addresses these limitations by implementing an integrated data model and interactive dashboards**,** enabling scalable, dynamic, and comprehensive economic analysis.

4. Source of Dataset

To achieve the objectives of this analysis, data was sourced from authorized government and open-data repositories related to India's economic performance. The datasets were selected to ensure reliability, accuracy, and relevance for analyzing GDP trends, city-level economic contribution, sectoral performance, and employment indicators.

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1. GDP and Economic Indicators Data

- **Source Description:**
  Official macroeconomic data published by Indian government institutions and internationally recognized economic repositories, providing reliable measurements of India's economic growth.
- **Data Sources:**
  - Ministry of Statistics and Programme Implementation (MoSPI), Government of India
  - Reserve Bank of India (RBI) – Economic Statistics
  - World Bank Open Data (India-specific indicators)
- **Key Attributes:**
  Year, GDP (Current and Constant Prices), GDP Growth Rate (Annual %), Sector-wise GDP Contribution.

- **Format:**
  Excel Workbooks (.xlsx) and CSV files.

2.City Level GDP Data

- **Source Description:**
  City-wise and urban economic estimates derived from government reports, planning commission publications, and authorized statistical studies related to urban economic output.
- **Key Attributes:**
  City Name, State, City GDP Contribution, Growth Rate, Sectoral Distribution.
- **Format:**
  CSV and Excel (.xlsx) files.

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3. Employment and Unemployment Data

- **Source Description:**
  Employment-related indicators sourced from official labor and economic databases to study workforce trends in relation to GDP growth.
- **Data Sources:**
  - Ministry of Labour and Employment, Government of India
  - Periodic Labour Force Survey (PLFS)
  - World Bank Open Data (India unemployment indicators)
- **Key Attributes:**
  Year, Employment Rate, Unemployment Rate, Workforce Participation Rate.

- **Format:**
  Excel Workbooks (.xlsx) and CSV files.

- **Data Constraints and Assumptions:** City-level GDP figures are based on estimates and aggregated reports, as official city-wise GDP data is not uniformly published every year.
- All datasets were aligned temporally and standardized to ensure consistency during analysis and modeling.

5. Data Preprocessing/ Cleaning

Data preprocessing is a crucial phase in the Business Intelligence lifecycle, as it ensures that raw economic data is transformed into a clean, accurate, and consistent format suitable for analysis and visualization. In this project, Power Query Editor in Microsoft Power BI was extensively used to perform all data cleaning and transformation operations.

The following key preprocessing steps were executed:

- **Data Profiling and Type Casting:**
- Initial inspection of the imported datasets revealed inconsistencies in data types across different sources. Columns such as Year, GDP values, GDP Growth Rate, Employment Rate, and Unemployment Rate were verified and converted to appropriate numerical formats to enable accurate calculations and aggregations. Date-related fields were standardized to ensure compatibility with time-based analysis and trend visualization.

- **Handling Missing Values (Nulls):**

  Some datasets contained missing or incomplete values, particularly in city-level GDP estimates and employment indicators.

- Records with critical missing values were carefully filtered or handled using conditional logic.
- For aggregate visualizations such as total GDP or average growth rate, null values were either excluded or replaced with appropriate defaults, depending on analytical relevance.
- This approach ensured that missing data did not distort overall trends or insights.

- **Standardization of Categorical Data:**

Categorical fields such as City Name, State Name, and Sector Name contained inconsistencies due to variations in spelling or naming conventions.

- Power Query's Replace Values, Trim, and Grouping functions were used to standardize entries.
- Economic sectors were grouped into broader categories such as Information Technology, Manufacturing, Services, Tourism, and MSMEs to simplify analysis and improve dashboard readability.

- **Creating Calculated Columns for Modelling:**

To support effective data modeling and time-based analysis, additional calculated columns were created:

- A Year column was derived from date-related fields where necessary.
- Growth-related columns, such as Year-on-Year GDP Growth, were generated to support comparative analysis.

These calculated columns played a vital role in establishing relationships between datasets during the data modeling phase.

Outcomes of the PreProcessing

As a result of these preprocessing steps, the final dataset achieved:

- Improved data consistency
- Reduced redundancy and errors
- Enhanced compatibility for relational modeling and visualization

This clean and structured data foundation enabled accurate, efficient, and insightful analysis within the Power BI environment.

6. Data Modelling

Although the source datasets used in this project were primarily flat files (CSV and Excel), effective and scalable analysis in Microsoft Power BI requires a well-structured data model. To achieve optimal performance, accurate filtering, and meaningful relationships across datasets, this project employs a Star Schema data modeling approach.

In this model, fact tables store quantitative economic measures such as GDP values, growth rates, and employment figures, while dimension tables provide descriptive context such as city, sector, and time. This separation ensures improved query efficiency and consistent slicing and filtering across all dashboard visuals.

Fact Tables:

- **GDP Fact Table:**

Contains numerical measures including GDP values, GDP growth rate, and sector-wise contribution across different years.

- **Employment Fact Table:**

Stores employment-related metrics such as employment rate, unemployment rate, and workforce participation indicators.

- **Dimension Tables: Date Dimension:**
  Includes Year and other time attributes, enabling time-based filtering and trend analysis.
- **City Dimension:**
  Contains city and state information, supporting city-level and regional comparisons.
- **Sector Dimension:**
  Defines economic sectors such as Information Technology, Manufacturing, Services, Tourism, and MSMEs, enabling sector-wise analysis.

Relationships: One-to-Many relationships were established between dimension tables and fact tables.

 The Date dimension serves as a common key to link economic indicators across multiple years.

 City and Sector dimensions ensure consistent filtering across all visuals and dashboard pages.

Benefits of the Data Model

- Improved query performance
- Accurate cross-filtering between visuals
- Reduced data redundancy
- Enhanced scalability for future dataset expansion

This structured data modeling approach forms the backbone of the Power BI dashboards and enables reliable, interactive analysis of India's economic performance.

7. Analysis on Dataset

This section presents the key analytical dashboards developed in this project to explore India's economic performance across multiple dimensions, including GDP trends, city-wise contribution, sectoral distribution, and employment indicators.

Analysis 1: National & City-Level GDP Overview

ii. Introduction

The "National & City-Level GDP Overview" dashboard provides a macro-to-micro perspective of India's economic structure. It is designed to enable users to analyze overall GDP trends while simultaneously drilling down into city-wise economic contributions. This dashboard highlights how major metropolitan cities drive national GDP and how their performance changes over time.

The analysis primarily focuses on identifying:

- Long-term GDP growth patterns
- Top contributing cities to India's economy
- Year-wise variations in economic output

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iii. General Description

This dashboard presents a consolidated view of India's GDP using:

- **KPI cards** displaying total GDP and growth rate
- **Line charts** showing GDP trends over multiple years
- **Bar charts** ranking cities by GDP contribution

- **Geographical visualizations** to illustrate regional economic dominance

Interactive slicers allow users to filter the data by year, city, and sector, enabling dynamic exploration and comparison.

iv. Specific Requirements, functions and formulas

To develop the interactive visualizations used in the India GDP & City-Level Economic Analysis dashboard**,** several DAX (Data Analysis Expressions**)** measures were created. These measures form the computational backbone of the dashboard and enable accurate aggregation, comparison, and trend analysis of economic indicators.

Key DAX Measures:

- **Total GDP:** This base measure calculates the total Gross Domestic Product for the selected filters (year, city, or sector) and is used across multiple visuals.

Code snippet

- Total GDP = SUM(GDP_Fact[GDP_Value])
- **GDP Growth Rate % :** This measure calculates the average GDP growth rate, enabling year-wise and comparative trend analysis.

Code snippet

```
GDP Growth Rate =
AVERAGE( GDP_Fact[GDP_Growth_Rate] )
```

- **City Wise GDP Contribution:** This measure computes the percentage contribution of a city to national GDP, helping identify dominant and emerging urban economies.

Code snippet

```
City GDP Contribution (%) =
```

```
DIVIDE(
    [Total GDP],
    CALCULATE( [Total GDP], ALL( City_Dim ) )
)
```

v. Analysis results

The analysis reveals clear and meaningful patterns in India's economic performance across time, cities, and sectors. The GDP by Year trend highlights a steady long-term growth trajectory, with a noticeable slowdown during the 2020 pandemic period, followed by a gradual recovery in subsequent years. This pattern reflects the broader impact of global disruptions and the resilience of India's economy during the recovery phase.

The city-wise GDP contribution analysis shows that economic output is highly concentrated in major metropolitan regions. Cities such as Mumbai, Delhi, Bengaluru, Chennai, and Hyderabad emerge as dominant contributors, collectively accounting for a significant share of national GDP. This concentration underscores the critical role of urban centers in driving India's economic growth.

Sectoral analysis further indicates that Information Technology and Services sectors contribute the largest portion of GDP, particularly in technology-driven cities. In contrast, Manufacturing and Tourism exhibit more varied performance across regions, reflecting differences in infrastructure, investment, and regional specialization.

The employment-related analysis demonstrates a strong relationship between GDP growth and employment levels**.** Cities with higher GDP contribution generally display better employment indicators and lower unemployment rates, highlighting the positive impact of economic expansion on job creation.

Overall, the results emphasize the importance of city-level and sectoral analysis in understanding India's economic structure, as national aggregates alone do not fully capture regional disparities and growth dynamics.

vi. Visualization

The visualization strategy was designed to ensure clarity, interpretability, and effective insight generation**:**

- **Line charts** were used to represent GDP and growth trends over time, clearly highlighting periods of decline and recovery.
- **Bar charts** enabled direct comparison of GDP contribution across cities and sectors, making dominant contributors easily identifiable.
- **Donut and stacked charts** were utilized to illustrate sectoral composition and relative contribution.
- **Geographical map visuals** provided spatial context, helping users visually identify economically significant regions across India.

This combination of visual techniques ensures that complex economic data is presented in an intuitive and interactive manner, allowing users to explore patterns, compare regions, and derive actionable insights efficiently.

Analysis 2: Sectoral Contribution Analysis

ii. Introduction

The Sectoral Contribution Analysis dashboard focuses on understanding how different economic sectors contribute to India's GDP at both the national and city levels. Since India's growth is driven by a diverse mix of industries, analyzing sector-wise performance is essential to identify economic strengths, regional specialization, and sectoral dependency across urban centers.

This analysis enables users to explore the contribution of key sectors such as Information Technology, Manufacturing, Services, Tourism, and MSMEs, and examine how sectoral dominance varies across cities and time periods.

iii. General Description

The dashboard presents a comprehensive view of sectoral performance using multiple visual elements, including:

- **Donut and stacked charts** showing percentage contribution of each sector to GDP
- **Bar charts** comparing sectoral GDP across major cities
- **Slicers** for year, city, and sector to enable dynamic filtering

By interacting with these visuals, users can drill down into specific cities to understand which sectors are driving local economic growth and how sectoral composition changes over time.

iv. Analysis results

The analysis reveals that the Services and Information Technology sectors contribute the largest share to India's GDP, particularly in metropolitan cities such as Bengaluru, Hyderabad, and Pune, where technology-driven industries dominate economic output.

The Manufacturing sector shows strong contribution in cities with established industrial infrastructure, such as Chennai, Ahmedabad, and Pune, highlighting regional specialization in production and industrial activity. Meanwhile, the Tourism sector plays a significant role in select cities, contributing to GDP through hospitality, travel, and related services.

The MSME sector emerges as a crucial supporting pillar of the economy, contributing consistently across cities by generating employment and supporting local economic activity. However, its GDP contribution varies based on access to infrastructure, finance, and urban development.

Overall, the results indicate that India's economic growth is sectorally diversified, but heavily influenced by urban centers where service-oriented and technology-driven industries are concentrated.

v. Visualization

- **Donut charts** were used to represent sectoral share of GDP, allowing quick identification of dominant sectors.
- **Stacked bar charts** enabled comparison of sector-wise contribution across multiple cities.
- **Interactive slicers** ensured flexible analysis by year and region.

This visualization approach helps users clearly understand sectoral dominance, regional economic specialization, and shifts in contribution patterns over time.

Analysis 3: Employment and Unemployment Analysis

ii. Introduction

The Employment and Unemployment Analysis dashboard focuses on examining the relationship between economic growth and workforce trends across major Indian cities. Employment indicators serve as a critical measure of economic well-being, as sustained GDP growth is expected to generate job opportunities and reduce unemployment levels.

This analysis aims to evaluate how GDP performance at the city and sectoral levels influences employment outcomes, and to identify regions where economic growth has translated effectively into job creation.

iii. General Description

This dashboard presents employment-related insights using a combination of:

- **KPI cards** displaying employment rate and unemployment rate
- **Line charts** illustrating employment and unemployment trends over time
- **Bar charts** comparing employment indicators across cities
- **Interactive slicers** for year and city selection

These visual elements allow users to dynamically explore how workforce trends evolve over time and how they differ between high-growth and moderate-growth urban regions.

v. Analysis results

The analysis indicates a strong correlation between GDP growth and employment levels across major Indian cities. Cities with higher GDP contribution, such as Mumbai, Bengaluru, Delhi, and Chennai**,** generally exhibit higher employment rates and lower unemployment levels, reflecting the availability of diverse economic opportunities.

During periods of economic slowdown, particularly around the 2020 pandemic phase, a temporary decline in employment rates and a rise in unemployment were observed across most cities. However, the recovery phase demonstrates that cities with strong service-oriented and technology-driven sectors rebounded more quickly in terms of job creation.

The analysis also highlights regional disparities in employment outcomes. While metropolitan cities show relatively stable workforce indicators, smaller or emerging cities exhibit greater variability in employment and unemployment trends, suggesting differences in economic resilience and industrial diversification.

Overall, the findings reinforce the importance of sustainable economic growth and sectoral diversification in maintaining stable employment levels across urban India.

vi. Visualization

- **Line charts** were used to clearly depict employment and unemployment trends over time.
- **Bar charts** enabled comparison of employment indicators across cities.
- **KPI cards** provided quick summaries of key workforce metrics.

This visualization strategy ensures that employment-related data is presented in a clear, interpretable, and decision-oriented manner.

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8. List of Analysis with results

The following table summarizes the key analytical findings derived from the Power BI dashboards developed in this project:

Analysis Component	Key Metric / Visual	Primary Finding / Result
GDP Overview	GDP by Year (Line Chart)	India's GDP shows steady long-term growth with a noticeable dip during the 2020 pandemic period, followed by gradual recovery in subsequent years.
City-Level Analysis	City-wise GDP Contribution (Bar Chart)	Major metropolitan cities such as Mumbai, Delhi, Bengaluru, Chennai, and Hyderabad contribute a significant share of national GDP, highlighting urban economic concentration.
Sectoral Analysis	Sector-wise GDP Contribution (Donut Chart)	The Services and Information Technology sectors contribute the largest share of GDP, particularly in technology-driven cities.
Sectoral Analysis	Sector GDP by City (Stacked Bar Chart)	Manufacturing dominates in industrial cities, while IT and services lead in metro and tech hubs, reflecting regional specialization.
Employment Analysis	Employment Rate by City (Bar Chart)	Cities with higher GDP contribution generally exhibit higher employment rates and stronger workforce participation.
Employment Analysis	Unemployment Trend (Line Chart)	Unemployment levels increased during economic slowdown phases but improved during periods of GDP recovery, indicating a strong GDP–employment relationship.
Comparative Analysis	GDP vs Employment (Combined View)	A positive correlation exists between economic growth and employment generation across major Indian cities.

9. Future scope

While this project delivers valuable insights into India's economic structure and urban growth patterns, several enhancements can be considered for future development:

1. **Real-Time Data Integration:**
   Future versions of the dashboard can **integrate** live economic data feeds from government portals or APIs to ensure real-time GDP and employment monitoring.

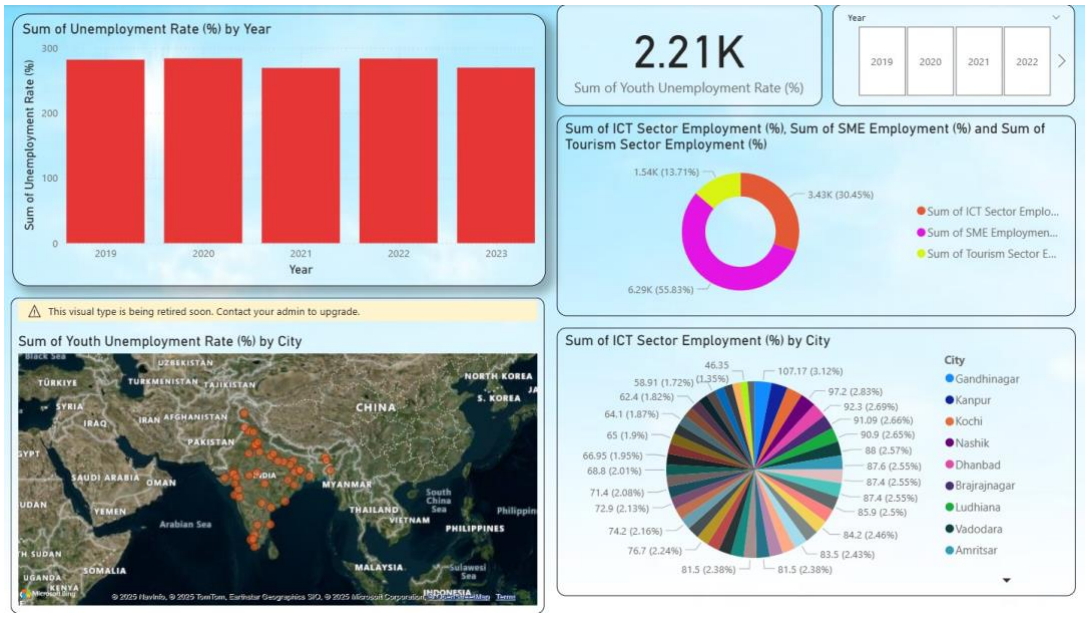
2. **Predictive Economic Forecasting:**
   Advanced forecasting techniques using Python within Power BI (such as ARIMA or regression models) can be implemented to predict future GDP growth and employment trends.

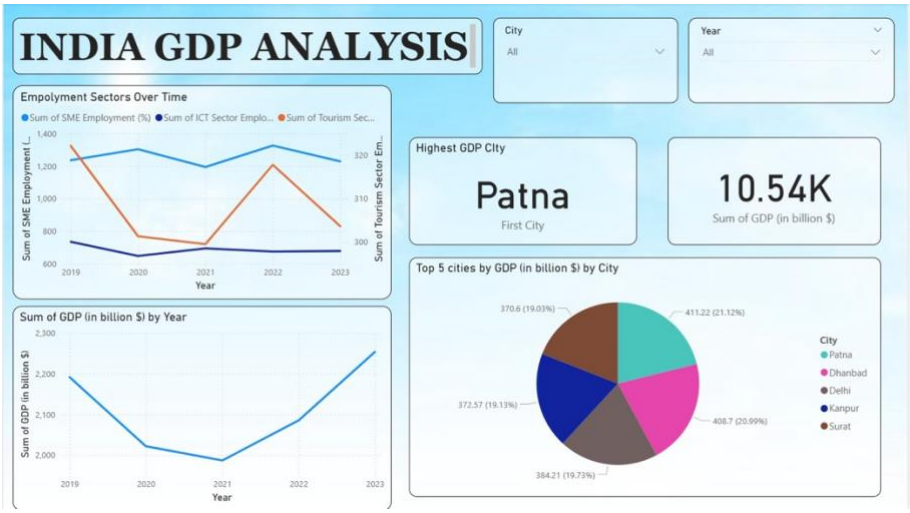
3. **Deeper Regional Analysis:**
   The analysis can be extended to include district-level or state-level GDP data, enabling more granular regional comparisons.

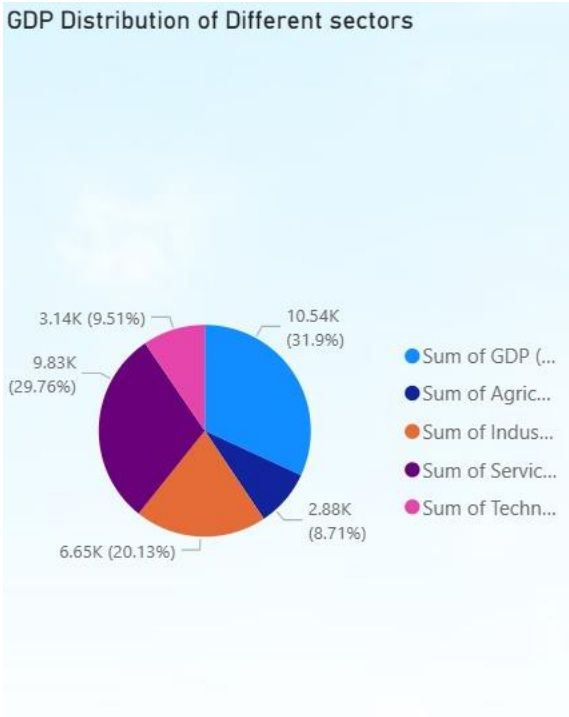
4. **Policy Impact Assessment:**
   Incorporating policy-related indicators (such as infrastructure spending or industrial incentives) can help assess the impact of government initiatives on regional economic growth.

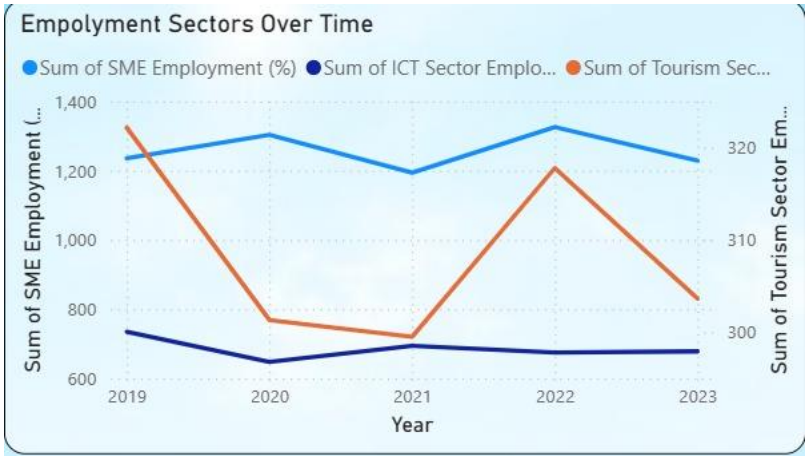
5. **Enhanced Employment Insights:**
   Future datasets may include sector-wise employment data, enabling a deeper understanding of job creation patterns across industries

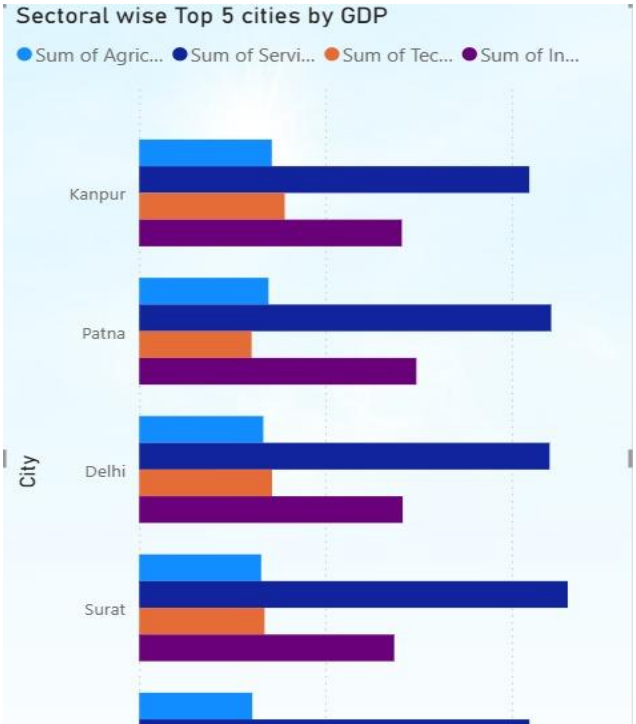
10.Images of the Dashboard

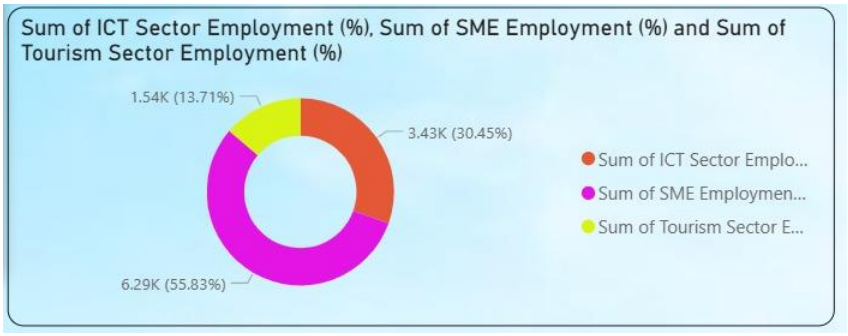

Sum of Unemployment Rate (%) by Year
Sum of Unemployment Rate (%)
300
200
100
0
2019
2020
2021
2022
2023
Year
2.21K
Sum of Youth Unemployment Rate (%)
Year
2019
2020
2021
2022
Sum of ICT Sector Employment (%), Sum of SME Employment (%) and Sum of Tourism Sector Employment (%)
1.54K (13.71%)
3.43K (30.45%)
6.29K (55.83%)
Sum of ICT Sector Emplo...
Sum of SME Employmen...
Sum of Tourism Sector E...
This visual type is being retired soon. Contact your admin to upgrade.
Sum of Youth Unemployment Rate (%) by City
Sum of ICT Sector Employment (%) by City
City
Gandhinagar
Kanpur
Kochi
Nashik
Dhanbad
Brajrajnagar
Ludhiana
Vadodara
Amritsar

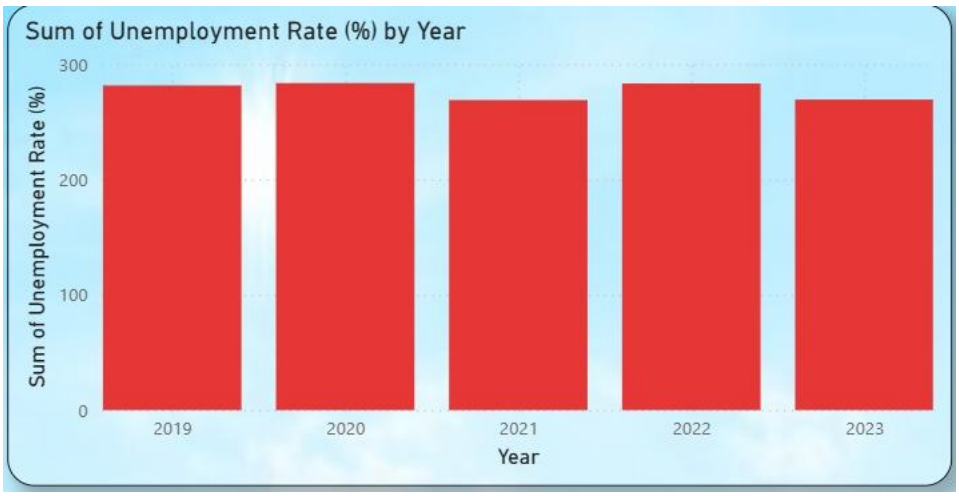

INDIA GDP ANALYSIS
City
All
Year
All
Empolyment Sectors Over Time
Sum of SME Employment (%)
Sum of ICT Sector Emplo...
Sum of Tourism Sec...
Sum of SME Employment (...
Sum of Tourism Sector Em...
Year
Highest GDP City
Patna
First City
10.54K
Sum of GDP (in billion $)
Sum of GDP (in billion $) by Year
Sum of GDP (in billion $)
Top 5 cities by GDP (in billion $) by City
370.6 (19.03%)
411.22 (21.12%)
372.57 (19.13%)
408.7 (20.99%)
384.21 (19.73%)
City
Patna
Dhanbad
Delhi
Kanpur
Surat


GDP Distribution of Different sectors
3.14K (9.51%)
10.54K (31.9%)
9.83K (29.76%)
2.88K (8.71%)
6.65K (20.13%)
Sum of GDP (...
Sum of Agric...
Sum of Indus...
Sum of Servic...
Sum of Techn...


Empolyment Sectors Over Time
Sum of SME Employment (%)
Sum of ICT Sector Emplo...
Sum of Tourism Sec...
Sum of SME Employment (...
1,400
1,200
1,000
800
600
Sum of Tourism Sector Em...
320
310
300
2019
2020
2021
2022
2023
Year


Sectoral wise Top 5 cities by GDP
Sum of Agric...
Sum of Servi...
Sum of Tec...
Sum of In...
City
Kanpur
Patna
Delhi
Surat


Sum of ICT Sector Employment (%), Sum of SME Employment (%) and Sum of Tourism Sector Employment (%)
1.54K (13.71%)
3.43K (30.45%)
6.29K (55.83%)
Sum of ICT Sector Emplo...
Sum of SME Employmen...
Sum of Tourism Sector E...


Sum of Unemployment Rate (%) by Year
Sum of Unemployment Rate (%)
300
200
100
0
2019
2020
2021
2022
2023
Year

This visual type is being retired soon. Contact your admin to upgrade.
Sum of Youth Unemployment Rate (%) by City
UZBEKISTAN
TÜRKIYE
TURKMENISTAN
TAJIKISTAN
NORTH KOREA
S. KOREA
SYRIA
IRAQ
IRAN
AFGHANISTAN
CHINA
PAKISTAN
SAUDI ARABIA
OMAN
MYANMAR
YEMEN
Arabian Sea
THAILAND
VIETNAM
South China Sea
PHILIPPINES
SOMALIA
MALAYSIA
Sulawesi Sea
UGANDA
KENYA
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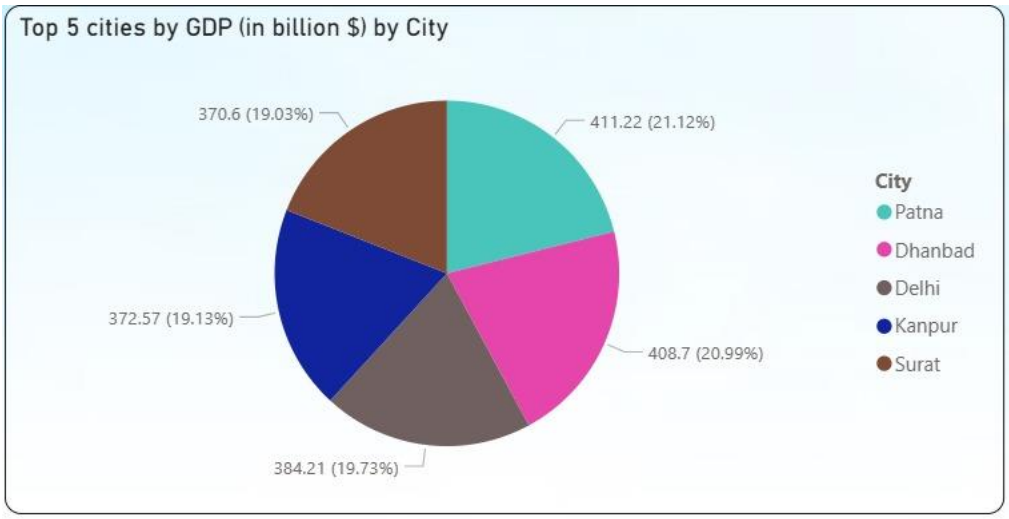
Top 5 cities by GDP (in billion $) by City
370.6 (19.03%)
411.22 (21.12%)
372.57 (19.13%)
408.7 (20.99%)
384.21 (19.73%)
City
Patna
Dhanbad
Delhi
Kanpur
Surat

10. References

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